Architecture Diagram

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1. Overview

StudyTrack follows a three-tier architecture, separating the application into a presentation layer (the React web app), an application layer (Node.js + Express services), and a data layer (PostgreSQL). External integrations with Canvas and Google Calendar feed data into the application layer to support the app's core differentiator: automatic import of academic deadlines.

2. Architecture Diagram

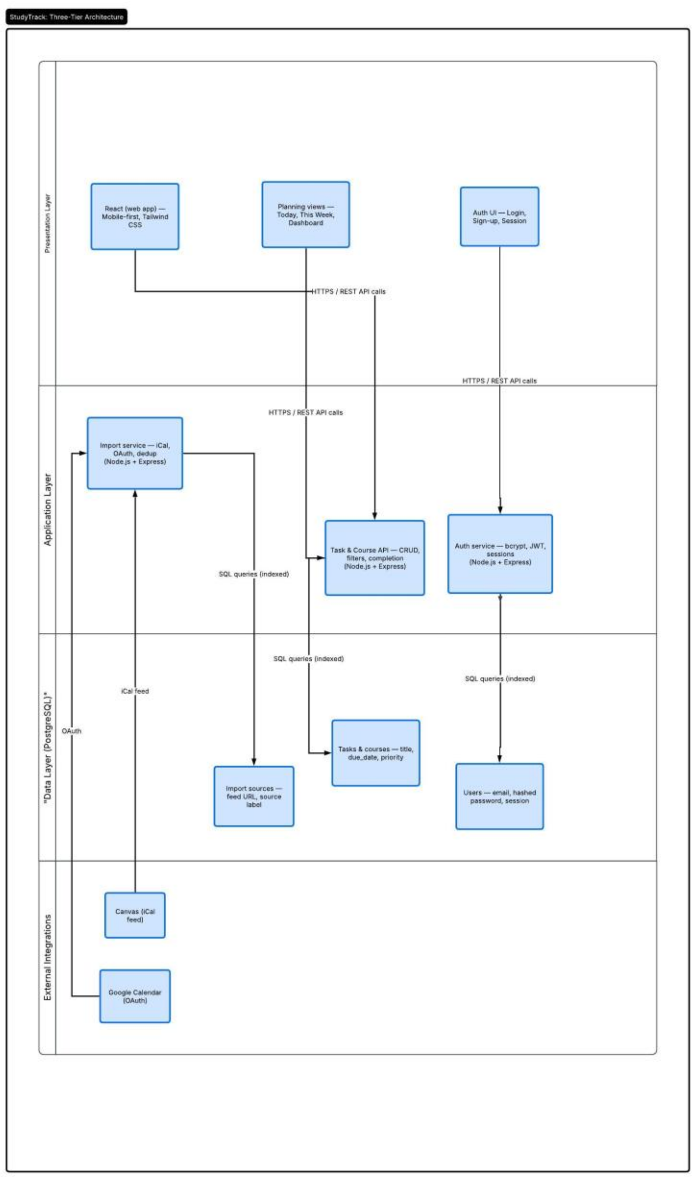

Figure 1: StudyTrack three-tier architecture

3. How It Works

3.1 Presentation Layer

The presentation layer is a mobile-first React web application built with Tailwind CSS. It provides three main surfaces: the authentication UI (login, sign-up, and session management), the planning views (Today, This Week, and the progress dashboard), and the overall React shell that ties the experience together. Every user action — creating a task, marking it complete, connecting a calendar — sends an HTTPS REST API request to the application layer.

3.2 Application Layer

Built on Node.js and Express, the application layer contains the system's business logic across three services:

- **Task & Course API —** handles CRUD operations for tasks and courses, applies filters (course, priority, completion status), and records task completion.
- **Auth service —** manages registration, login, and logout using bcrypt password hashing and session/JWT-based authentication.
- **Import service —** retrieves Canvas assignments via iCal feed and Google Calendar events via OAuth, converts them into tasks, maps them to the correct course, and de-duplicates on re-import.

3.3 Data Layer

All persistent data is stored in PostgreSQL across three logical groups:

- **Tasks & courses —** title, description, due date, priority, estimated effort, course association, and completion history.
- **Users —** email, hashed password, and session/token information.
- **Import sources —** feed URLs, source labels (Canvas or Google Calendar), and connection status, enabling refresh and de-duplication.

3.4 External Integrations

Canvas (via iCal feed) and Google Calendar (via OAuth) sit outside StudyTrack's own infrastructure. The Import service connects to these sources, pulls in assignment and event data on at least a daily basis, and converts new items into tasks without creating duplicates on subsequent syncs. Students can disconnect a source at any time and choose whether to keep or remove its previously imported tasks.

3.5 Data Flow Summary

A typical request flows from the React UI (presentation layer) to one of the Express services (application layer) via an authenticated HTTPS REST call. The service applies business logic and validation, then reads or writes data in PostgreSQL (data layer) using indexed SQL queries. Imported calendar data follows a separate path: the Import service periodically pulls from Canvas and Google Calendar, transforms the results into tasks, and writes them into the Tasks & courses and Import sources tables — after which they appear in the student's Today and This Week views like any manually created task.